

MONTCELO, AR, USA

WMO: 720175

Lat: 33.636N

Lon: 91.756W

Elev: 88

StdP: 100.27

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 53919

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.3	-3.1	-12.8	1.3	-1.5	-10.1	1.6	-0.6	8.3	14.8	7.6	14.0	1.7	40	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	36.9	25.2	35.4	24.8	34.0	24.8	26.9	33.0	26.4	32.6	25.9	31.8	2.7	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.4	20.8	29.7	24.8	20.0	29.3	24.1	19.2	28.8	84.8	32.8	82.5	32.6	80.3	32.0	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	6.1	5.4		-8.8	38.6	2.4	2.1	-10.6	40.1	-12.0	41.3	-13.3	42.5	-15.1	44.0	
			DB	-9.7	28.1	2.2	1.2	-11.3	29.0	-12.6	29.7	-13.9	30.3	-15.5	31.2	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.1	7.0	8.7	13.4	18.3	22.5	26.5	28.1	28.0	24.6	18.5	12.5	8.1	
	DBStd	8.82	6.06	5.81	5.38	4.40	3.51	2.35	2.34	2.50	3.51	4.74	5.09	5.51	
	HDD10.0	388	133	87	27	3	0	0	0	0	0	2	32	104	
	HDD18.3	1422	353	273	169	54	9	0	0	0	2	59	186	318	
	CDD10.0	3332	41	51	133	251	388	495	560	559	438	264	106	47	
	CDD18.3	1325	2	3	16	52	139	245	302	301	191	63	9	3	
	CDH23.3	13654	3	15	111	425	1211	2555	3452	3394	1863	568	53	6	
CDH26.7	6181	0	1	16	90	425	1179	1697	1741	842	185	7	0		
Wind	WSAvg	2.3	2.7	2.7	2.8	2.8	2.3	2.0	1.9	1.7	1.7	2.0	2.3	2.5	
Precipitation	PrecAvg	1366	124	119	134	137	138	93	82	77	77	114	111	148	
	PrecMax	1945	338	283	300	489	333	290	191	282	230	508	293	243	
	PrecMin	852	14	25	33	37	34	21	17	13	6	5	47	28	
	PrecStd	328	67	78	67	95	74	69	55	58	56	110	67	56	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.5	25.4	28.5	31.1	33.4	36.7	38.5	38.8	37.2	32.8	27.6	24.1	
		MCWB	17.5	17.8	19.1	21.2	23.6	24.0	24.6	25.2	24.1	21.5	19.7	19.8	
	2%	DB	21.3	22.8	26.4	28.9	32.0	34.9	36.3	37.2	34.9	30.9	24.9	21.6	
		MCWB	16.8	17.2	18.2	20.4	23.0	24.4	25.6	25.8	23.4	22.2	18.7	18.1	
	5%	DB	19.1	20.6	24.1	27.5	30.8	33.6	34.8	35.6	33.2	28.7	22.8	19.6	
		MCWB	15.6	16.3	17.0	19.5	22.6	24.4	25.5	25.3	23.5	20.8	17.7	16.4	
	10%	DB	17.1	18.5	22.2	26.1	29.1	32.4	33.5	33.9	31.6	26.9	21.0	17.4	
		MCWB	13.7	14.8	16.1	18.8	22.0	24.2	25.5	25.0	23.1	19.8	16.9	14.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.5	20.0	20.7	23.3	25.4	26.9	27.8	27.9	26.1	24.9	21.9	20.6
			MCDB	21.9	23.2	25.8	28.2	30.5	32.1	33.4	34.3	32.3	29.7	24.7	23.0
2%		WB	18.1	18.7	19.5	22.0	24.4	26.2	27.0	26.9	25.4	23.7	20.3	19.0	
		MCDB	20.3	21.4	24.3	26.5	29.3	31.6	32.9	33.5	31.4	28.2	23.3	21.1	
5%		WB	16.6	17.1	18.6	21.2	23.8	25.6	26.6	26.3	24.9	22.6	19.0	17.4	
		MCDB	18.6	19.8	22.8	25.5	28.6	31.0	32.5	32.9	30.4	26.5	21.9	18.9	
10%		WB	14.0	15.4	17.4	20.3	23.0	25.0	26.0	25.8	24.2	21.4	17.5	15.0	
		MCDB	16.7	18.4	21.0	24.4	27.8	30.4	31.8	31.9	29.2	25.0	20.0	17.1	
Mean Daily Temperature Range	5% DB	MDBR	10.2	10.4	11.3	11.7	10.9	10.5	10.1	11.0	11.8	12.2	11.3	10.2	
		MCDBR	11.9	12.7	12.8	12.6	11.9	12.0	11.4	12.6	13.3	13.6	12.8	11.5	
		MCWBR	8.4	8.3	6.7	5.8	4.8	4.3	3.7	4.0	4.4	6.1	7.6	8.4	
	5% WB	MCDBR	10.4	10.7	11.4	10.4	10.4	10.1	10.1	11.0	10.7	11.1	10.4	10.4	
		MCWBR	8.8	8.1	6.9	5.8	5.0	4.3	4.0	4.2	4.0	6.2	7.5	8.8	
Clear-Sky Solar Irradiance	taub	0.320	0.334	0.361	0.401	0.436	0.475	0.489	0.486	0.422	0.361	0.339	0.318		
	taud	2.522	2.477	2.419	2.300	2.260	2.175	2.169	2.168	2.330	2.476	2.493	2.537		
	Ebn at Noon	896	918	915	888	856	819	804	801	841	875	864	878		
	Edh at Noon	83	97	111	130	137	149	149	147	120	96	85	78		
All-Sky Solar Radiation	RadAvg	2.56	3.01	4.26	5.32	5.89	6.50	6.26	5.79	4.98	4.00	2.87	2.26		
	RadStd	0.23	0.51	0.37	0.35	0.41	0.46	0.35	0.47	0.42	0.53	0.32	0.25		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page